REVIEW

Culinary seed oils: lipid outcomes, clinical evidence, and heating-related exposures

REFERENCES	STYLE	MADE WITH	VERIFICATION	COMPILED
74 verified	Scientific	Grounded v0.5.2	Crossref · claims	5 Sep 2026

ABSTRACT

This narrative review examines lipid outcomes, clinical events, inflammatory pathways, and heating-related exposures associated with culinary seed oils. Replacing saturated fats with unsaturated oils lowers low-density lipoprotein cholesterol. Substitution trials provide some evidence of reduced coronary events, while broader and omega-6-specific analyses leave uncertainty about mortality. Recovered historical trials include adverse or null mortality findings; their mixed interventions and methodological limitations restrict applicability to current unhydrogenated oils. Reviews found little evidence that increasing linoleic acid raises common circulating inflammatory markers, but a newer controlled trial reported changes in fatty-acid concentrations and stimulated blood-cell mediator production. These biochemical responses have unresolved clinical implications. Confidence is moderate for short-term lipid effects and lower for disease prevention from particular oils. Heating alters oil chemistry, but the magnitude of risk at ordinary human dietary exposures remains very uncertain. Conclusions depend on the oil, comparator, outcome, and preparation conditions evaluated.

01 Introduction

The term "seed oils" encompasses oils with different nutrient compositions and preparation conditions. Analyses found linoleic-acid content spanning 1.6% to 79.0% of fatty acids across culinary oils.[1] A Nordic scoping review notes that an oil's effects on risk markers may vary with its polyphenols, phytosterols, and other minor components.[2] A nutritional comparison therefore requires information about oil composition as well as botanical origin.

This review examines how findings differ by oil, replacement food, population, preparation, and outcome. It considers five evidence categories: controlled changes in risk markers, randomized clinical events, prospective disease associations, mechanistic and inflammatory evidence, and degradation during heating. Human dietary interventions are evaluated separately from observational associations and laboratory experiments, because these designs address different aspects of exposure and disease.

The assessment distinguishes effects on low-density lipoprotein (LDL) cholesterol from effects on clinical events and mortality. It assigns moderate confidence to short-term lipid effects of dietary replacement and lower confidence to disease prevention attributable to particular seed oils. These are narrative judgments about the relevant evidence bodies; their basis includes study design, consistency, and applicability to the exposure being considered.

02 Oil composition and exposure definition

Vegetable oils vary in fatty-acid composition and minor components.[2] An umbrella review reported different health benefits among vegetable oils, with lipid reductions especially for oils rich in monounsaturated and polyunsaturated fats.[3] The Nordic scoping review reached a narrower conclusion: food-level oil syntheses were sparse and generally of low methodological quality.[2] The limited food-level evidence restricts comparisons between named oils and between oils consumed in different dietary contexts.

Linoleic and arachidonic acids have distinct physiological roles within omega-6 biology.[4] Some mediators derived from the omega-6 fatty acid arachidonic acid are mainly pro-inflammatory, although ordinary dietary linoleic- and arachidonic-acid intake has shown little effect on inflammatory biomarkers.[4] Consequently, high-linoleic and high-oleic sunflower oils, corn and canola oils, and partially hydrogenated historical margarines should be specified separately when interpreting intervention results.

Figure 1 distinguishes dietary comparisons with blood measurements from clinical follow-up.

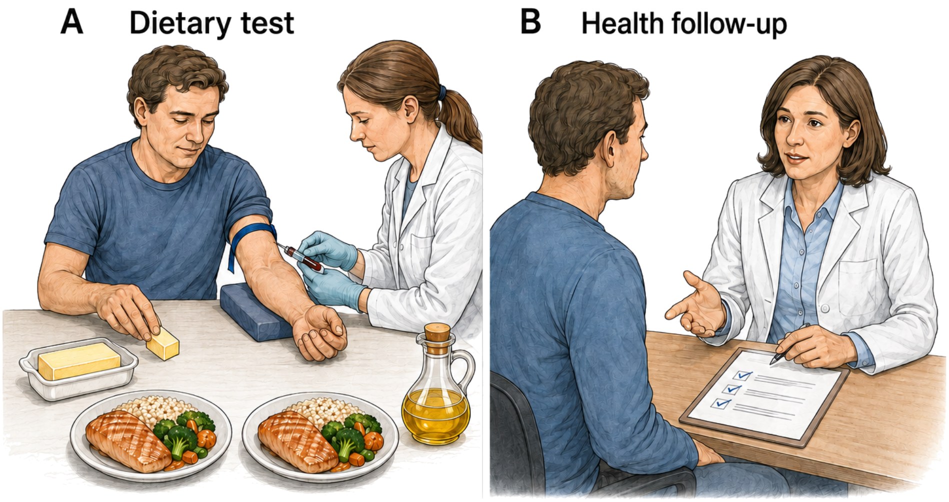

Figure 1. Dietary comparisons and clinical outcomes. Dietary substitution trials measure lipid responses to replacement of saturated fats with unsaturated oils.[5] Clinical-event syntheses evaluate different endpoints and retain uncertainty for several omega-6-specific outcomes.[6] Panel A schematically depicts a dietary comparison and blood collection; panel B depicts clinical follow-up. The meals and consultations are explanatory illustrations, not reconstructions of an individual trial or evidence that a lipid change guarantees a clinical benefit.

03 Lipid outcomes in dietary substitution trials

A systematic review found convincing evidence that replacing saturated fat with polyunsaturated or monounsaturated fat lowers total and LDL cholesterol.[5] A review focused on linoleic acid reached the same directional conclusion for lipid risk markers in healthy adults.[7]

A network meta-analysis of 54 trials reported LDL-cholesterol differences ranging from −0.42 to −0.23 mmol/L per isocaloric replacement of 10% of dietary energy from butter with oils.[8] This is a range across comparisons with butter, rather than a confidence interval for a single oil.

Trials comparing named oils provide more specific estimates. In a crossover trial, corn oil produced a more favorable lipid profile than coconut oil over four weeks.[9] Replacing dairy fat with rapeseed oil lowered serum cholesterol, triglycerides, and LDL cholesterol in adults with hyperlipidaemia.[10] A soybean-oil mayonnaise lowered total and LDL cholesterol relative to palm-olein mayonnaise, while leaving oxidative-stress markers unchanged.[11] Products low in saturated and trans fatty acids also improved lipoprotein cholesterol relative to more hydrogenated products.[12] These trials support comparisons of short-term lipid responses, without establishing an effect on myocardial infarction.

A review reported lower LDL cholesterol with coconut oil than with butter, but did not report a corresponding advantage over cis-unsaturated vegetable oils.[13] In a randomized three-way comparison, LDL-cholesterol outcomes with coconut oil were more similar to those with olive oil than to those with butter.[14] A systematic review of high-oleic oils found comparable lipid benefits when saturated or trans fats were replaced by high-oleic or high-polyunsaturated oils.[15]

04 Cardiovascular events and mortality in randomized trials

Clinical-event syntheses are less uniform than lipid trials. A Cochrane review of increased polyunsaturated fat found little or no effect on all-cause mortality but a probable small reduction in coronary and cardiovascular events.[16] A separate meta-analysis of eight randomized trials, 13,614 participants, and 1,042 coronary events estimated a 19% event reduction when polyunsaturated fat replaced saturated fat (risk ratio [RR] 0.81, 95% confidence interval 0.70–0.95).[17] The estimated reduction applies to replacement of saturated fat with polyunsaturated fat (PUFA).

A review limited to omega-6 interventions judged the evidence less certain. It found little or no difference in all-cause mortality or cardiovascular events, with wide intervals for coronary outcomes.[6] Ramsden's analysis distinguished mixed omega-3/omega-6 interventions from omega-6-specific trials.[18] Thus a linoleic-acid-specific question requires a narrower interpretation than a question about total polyunsaturated fat. Confidence in a small reduction in coronary events is higher than confidence in a mortality reduction.

Because review overlap was not fully reconstructed, these syntheses were not treated as independent replications. The combined assessment supports a possible reduction in coronary events with particular replacement regimens, with lower confidence in attributing that reduction specifically to omega-6-rich oils. Broad PUFA interventions and omega-6-

specific interventions are retained as separate comparisons because they do not define the same exposure.

The clinical contrasts are kept separate in the following table.

SYNTHESIS	INCLUDED CONTRAST	CORONARY ESTIMATE	MORTALITY CONCLUSION	SOURCE
Broad PUFA review	Higher total PUFA	Coronary events probably slightly lower	Mortality little changed	16
Substitution trials	PUFA for saturated fat	RR 0.81 (95% interval 0.70–0.95)	Coronary endpoint	17
Omega-6 review	Higher omega-6	Coronary effect uncertain	Mortality RR 1.00 (0.88–1.12)	6

Figure 2 presents the coronary-event estimates and their confidence intervals from the separate syntheses.

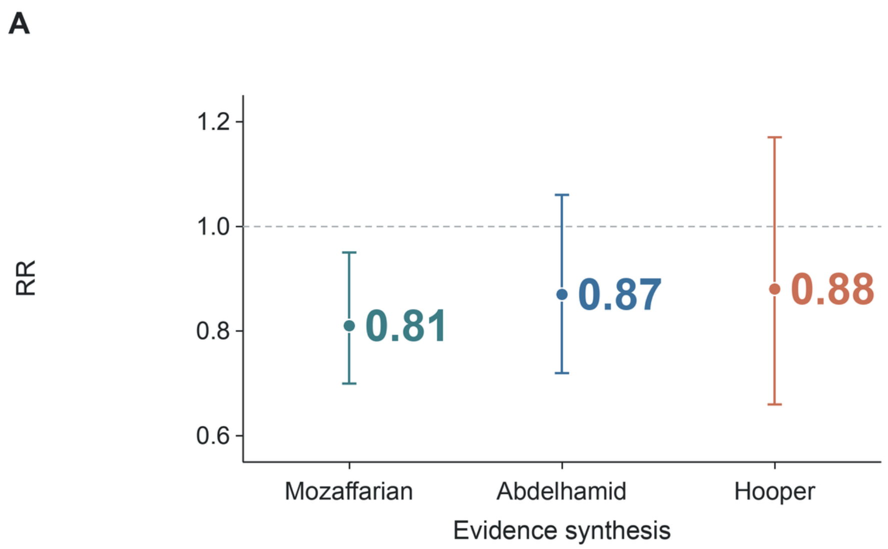

Figure 2. Coronary-event risk ratios from randomized-trial syntheses. A substitution-focused synthesis reported a risk ratio of 0.81 (95% confidence interval 0.70–0.95).[17] A broader polyunsaturated-fat synthesis reported 0.87 (0.72–1.06).[16] An omega-6-specific synthesis reported 0.88 (0.66–1.17).[6] Their visual alignment is not a new pooled estimate. Vertical position gives coronary risk ratios; horizontal categories identify syntheses. Whiskers show confidence intervals; one means no difference.

05 Historical diet–heart trials and intervention composition

Reanalyses of historical diet–heart trials reported adverse or null mortality results. In the Sydney Diet Heart Study, 458 men with recent coronary events were assigned to a safflower-oil intervention or usual care.[19] The intervention group had higher all-cause mortality (hazard ratio [HR] 1.62, 95% confidence interval 1.00–2.64) and cardiovascular mortality (1.70, 1.03–2.80).[19] A reanalysis separating omega-6-specific from mixed-polyunsaturated interventions similarly reported no demonstrated benefit and a possible adverse direction for omega-6-specific trials.[18] Their applicability depends on the intervention composition and the populations studied.

The Minnesota Coronary Experiment also evaluated a linoleic-acid-rich intervention. Replacing saturated fat with linoleic acid lowered serum cholesterol without a demonstrated reduction in coronary or all-cause mortality.[20] The original Minnesota report likewise found no difference in cardiovascular events, cardiovascular deaths, or total mortality in the overall population.[21] A stricter meta-analysis later argued that apparent benefit depended on including inadequately controlled trials.[22] Cholesterol reductions in these studies were not accompanied by demonstrated mortality reductions.

The historical interventions included foods that differ from current unhydrogenated liquid oils. The Sydney intervention included safflower-oil margarine as well as liquid oil.[19] The Minnesota diets used corn-oil polyunsaturated margarine against control margarines and shortenings.[20] Reviews of partially hydrogenated oils show that expected coronary effects depend on both trans-fat content and the replacement oil.[23] Because these interventions changed several dietary fat sources, their results apply to the historical regimens and do not isolate the effect of a current unhydrogenated liquid oil.

These mortality findings should be considered alongside the coronary-event results from broader substitution analyses. A broader review estimated about 10% lower coronary risk when 5% of energy from saturated fat was replaced by polyunsaturated fat, but found mixed evidence for several other cardiometabolic outcomes.[24] Differences in interventions, control quality, and outcomes limit direct comparison between the historical trials and broader evidence syntheses.

06 Prospective fatty-acid biomarkers and cardiovascular outcomes

Prospective biomarker studies include inverse and null associations with cardiovascular outcomes. In pooled international analyses, higher circulating or tissue linoleic acid was associated with lower total cardiovascular disease, cardiovascular mortality, and ischemic stroke.[25] A later meta-analysis found that each standard-deviation increase in linoleic acid was associated with a 10% lower risk of fatal coronary disease, with no clear reduction in nonfatal disease.[26] In older adults, high circulating linoleic acid was associated with lower total and coronary mortality.[27] EPIC-Norfolk likewise found an inverse association between omega-6 phospholipid fatty acids and coronary disease.[28]

The Multi-Ethnic Study of Atherosclerosis found no significant association between omega-6 fatty acids and incident cardiovascular disease.[29] A Dutch cohort found that a 4–5 percentage-point energy difference in linoleic acid or carbohydrate intake was not associated with coronary incidence.[30] Adipose-tissue linoleic acid was not associated with myocardial infarction in a Danish case-cohort study.[31] Across eleven cohorts, linoleic- and arachidonic-acid biomarkers were not associated with incident atrial fibrillation.[32] The observed associations therefore differ by cardiovascular endpoint and study.

Genetic analyses provide a separate approach to evaluating causal interpretations of biomarker associations. A Mendelian randomization analysis using genetic-association data from more than 114,000 UK Biobank participants did not support a protective role of circulating polyunsaturated-fatty-acid concentrations against cardiovascular disease and identified horizontal pleiotropy as a potential bias.[33] The genetic exposure in this analysis was circulating fatty-acid concentration, rather than consumption of a named oil.

Figure 3 keeps the cardiovascular and diabetes outcomes in separately identified categories.

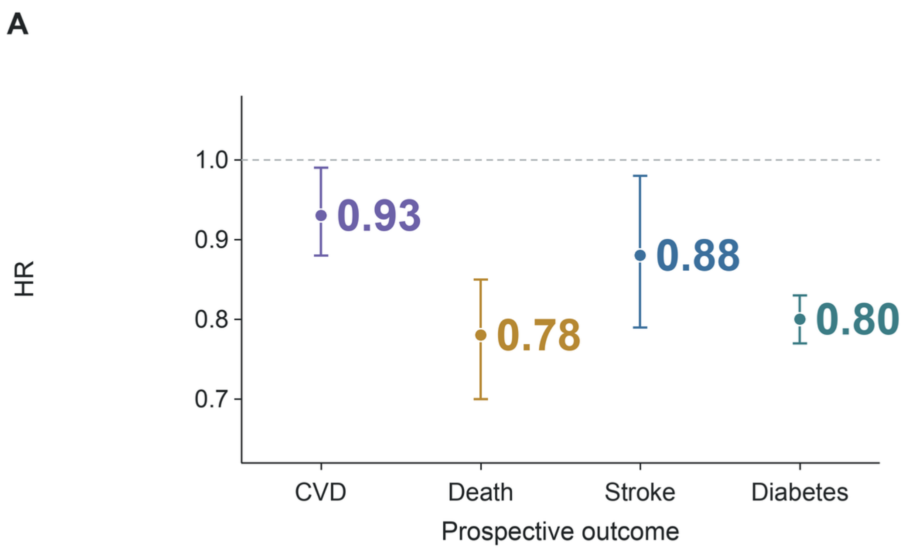

Figure 3. Prospective associations of circulating linoleic acid with cardiovascular outcomes and diabetes. Per interquintile-range increase in linoleic acid, pooled biomarker analyses reported HR 0.93 for total cardiovascular disease, 0.78 for cardiovascular mortality, and 0.88 for ischemic stroke.[25] EPIC-InterAct reported HR 0.80 per standard-deviation increase in linoleic acid for its association with type 2 diabetes.[34] Vertical position gives hazard ratios; horizontal categories identify outcomes. Whiskers show confidence intervals. Exposure contrasts differ; associations are not oil-intervention effects.

The assessment assigns low confidence to a causal cardiovascular benefit inferred from these biomarkers. Horizontal pleiotropy may limit the genetic analysis because the instruments can influence disease through other lipid-related routes.[33] Inverse observational associations and uncertain genetic estimates do not establish the effect of a specified dietary substitution on cardiovascular disease.

07 Type 2 diabetes and glucose regulation

In a pooled analysis of 39,740 adults from 20 cohorts, higher linoleic-acid biomarkers were associated with lower incident type 2 diabetes.[35] EPIC-InterAct also found linoleic acid inversely associated with diabetes, while several less abundant omega-6 fatty acids had different directions.[34]

The associations should be interpreted for the individual fatty acids measured.

Randomized evidence distinguishes diabetes diagnoses from intermediate glucose measures. A meta-analysis of omega-3, omega-6, and total polyunsaturated-fat trials found very uncertain effects of omega-6 on diabetes diagnosis and little or no effect on most measures of glucose metabolism.[36] By contrast, a meta-analysis of 102 controlled feeding trials found that replacing carbohydrate or saturated fat with polyunsaturated fat improved several measures of glucose–insulin regulation.[37] The findings concern different endpoints and durations. Improved glucose–insulin regulation in feeding trials does not by itself establish a reduction in incident diabetes.

08 Circulating inflammatory markers

Reviews of human interventions have evaluated changes in circulating inflammatory markers after increasing linoleic-acid intake. A systematic review of human interventions found virtually no evidence that adding linoleic acid increased common inflammatory markers in healthy adults.[38] A later meta-analysis also found no significant overall change in blood inflammatory markers when dietary linoleic acid increased.[39] A review noted that high dietary or circulating linoleic acid was not associated with stronger in-vivo or ex-vivo inflammatory responses.[40]

The assessment assigns moderate confidence to the findings for these measured circulating markers. Their interpretation does not extend to all inflammatory pathways, particularly responses to laboratory stimulation that differ from the circulating-marker endpoints considered in these reviews.

09 Lipid mediators and mechanistic evidence

Mechanistic reviews differ in their interpretation of omega-6-related inflammatory pathways. A review of omega-6 inflammation describes both adverse and neutral or favorable signals.[41] A mechanistic review emphasizes pro-inflammatory eicosanoids and associates a high omega-6-to-omega-3 ratio with chronic disease.[42] A discussion of plasma-fatty-acid findings reported associations with lower pro-inflammatory and higher anti-inflammatory markers.[43]

The omega-6-to-omega-3 ratio has also been proposed as a dietary target. A review argues that a lower omega-6-to-omega-3 ratio is broadly desirable.[44] A review of human studies found that few clinical studies tested the ratio directly and that their results conflicted.[45] A soybean-oil review reports that health agencies emphasize adequate intake of each fatty-acid family rather than a single ratio target.[46] A contrary review links seed-oil linoleic acid to oxidized metabolites and inflammatory pathways.[47] Interpretation of a ratio requires the underlying omega-6 and omega-3 intakes to be specified.

Figure 4 locates the biochemical measurement outside the participant.

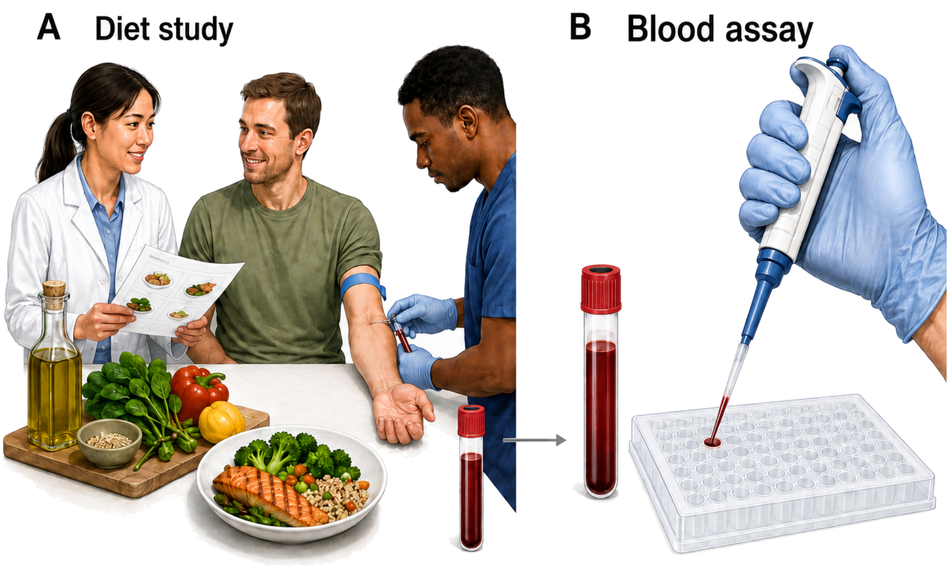

Figure 4. Dietary intervention and sampled-blood analysis. The dietary trial evaluated mediator production after laboratory stimulation of whole blood and did not directly assess clinical outcomes.[48] Panel A schematically depicts dietary guidance and blood collection; panel B depicts sample handling for an assay. The arrow denotes transfer of a sample, not a disease effect. The plate is a schematic laboratory representation; no assay result or molecular mechanism is encoded.

A narrative review links linoleic-acid oxidation products to oxidized LDL and atherosclerotic lesions.[49] Toxicology reviews show that aldehydic lipid-oxidation products can form, enter food, and interact with biological systems.[50] Yet the latter review explicitly calls for nutritional and epidemiological trials connecting quantified exposure to chronic disease.[50] The proposed mechanism is more directly relevant to degraded, heated oils than to ordinary unheated intake, for which the human exposure–outcome relationship remains uncertain.

Sergeant and colleagues compared high- and low-linoleic-acid diets for twelve weeks, using safflower-oil variants while holding alpha-linolenic-acid exposure similar.[48] Of 80 randomized participants, 52 completed the intervention.[48] The high-linoleic-acid arm had lower plasma eicosapentaenoic acid (EPA), an omega-3 fatty acid, and higher ratios of arachidonic-acid-derived to EPA-derived products after laboratory stimulation of whole blood.[48] The study therefore documented a biochemical response to the dietary comparison.

The clinical implications of that response remain uncertain. The authors explicitly state that the study does not directly address clinical outcomes.[48] They also describe substantial dietary support, no common-diet run-in, and a reciprocal increase in oleic acid when linoleic acid was reduced.[48] The assessment therefore gives moderate confidence to the observed biochemical contrast and much less confidence to any prediction of disease from it. Attrition and the concurrent oleic-acid substitution further limit attribution of the response to an isolated change in linoleic acid.

Further trials could assess whether the mediator changes persist and whether they predict clinical outcomes under specified replacement diets. Such studies would need to distinguish linoleic-acid effects from accompanying changes in other fatty acids.

10 Short-term comparisons of named oils

A meta-analysis found lower LDL cholesterol, total cholesterol, and the ratio of LDL to high-density lipoprotein (HDL) cholesterol with canola oil compared with olive oil.[51] In women with polycystic ovary syndrome, a ten-week trial reported lipid-profile improvements with canola compared with sunflower and olive oils.[52] In patients referred for coronary angiography, canola and olive oil produced different inflammatory-marker patterns without clear overall lipid differences.[53] A high-oleic canola intervention altered LDL composition without increasing LDL proteoglycan binding.[54] These outcomes support biochemical comparisons in the studied populations; they do not establish durable clinical benefits.

Corn oil improved atherogenic lipids relative to extra-virgin olive oil in a controlled feeding trial.[55] Corn oil produced a more favorable plasma lipid profile than coconut oil in another trial.[9] Palm oil raised LDL cholesterol relative to vegetable oils lower in saturated fat.[56] Hydrogenating linoleic acid toward trans or stearic fatty acids raised LDL and lowered HDL relative to linoleic acid.[57] A palm-oil crossover trial found only modest changes when palm replaced common dietary fats.[58] These results pertain to different formulations and replacement foods and should not be interpreted as a common comparison.

COMPARISON	DURATION	MAIN FINDING	SOURCE
Corn vs coconut	4 weeks	Better plasma lipid profile with corn	9
Soybean mayonnaise vs palm-olein mayonnaise	4 weeks	Lower total and LDL cholesterol with soybean mayonnaise	11
Canola vs olive	Trial meta-analysis	Lower LDL with canola	51

11 Cooking-oil intake and disease associations

In a cohort of 521,120 older US adults, corn and canola oil were not associated with higher cardiometabolic mortality per tablespoon, while modeled replacement of butter or margarine with nonhydrogenated vegetable oils was associated with lower mortality.[59] A Chinese cross-sectional analysis associated vegetable/sesame-oil use with higher prevalent atherosclerotic cardiovascular disease than lard or other animal-fat use.[60] Its cross-sectional design cannot establish whether oil choice preceded disease or the effect of changing oil use on subsequent cardiovascular incidence.

Dietary-pattern analyses provide additional context for the cooking-oil associations. An etiologic synthesis found that estimated effects of individual dietary components were similar to quantified effects of dietary patterns on cardiovascular risk factors and clinical endpoints.[61] A recent macronutrient meta-analysis found that total-fat associations changed when analyses were stratified by fat type.[62] Food-level associations therefore require interpretation in the context of replacement foods and the surrounding dietary pattern.

In three US health-professional cohorts, the highest total plant-oil intake was associated with lower total mortality than the lowest intake, with a hazard ratio of 0.84 and a 95% confidence interval of 0.79–0.90.[63] The analysis used repeated dietary questionnaires and modeled replacements of butter with plant oils.[63] This was an observational substitution analysis rather than a randomized comparison. The account here is limited to the estimates and design information available in the retrieved abstract.

The assessment retains low certainty for disease prevention inferred from food-level associations. Differences in exposure definitions and study design, together with the absence of randomized allocation to the modeled replacements, limit inference about the size of a causal effect.

12 Heating, storage, and lipid-oxidation products

In a controlled experiment, oxidation products rose from 100°C to 200°C, and several aldehydes were highest in polyunsaturated soybean oil at 200°C.[64] Oxidative stability fell about 30% across the tested refined oils after twelve months of storage.[65] Rapeseed-oil experiments detected tertiary degradation products at advanced stages of lipid oxidation.[66] Heating conditions are therefore part of the exposure definition.

Rapeseed oil incubated at 160°C for 120 minutes yielded up to 65 detected volatile compounds.[66] After 10 minutes, aged oil produced a 7-fold higher hexanal amount than non-aged oil.[66] These measurements describe compound detection and relative hexanal production under the experimental conditions.

Figure 5 separates an oil-heating experiment from clinical exposure assessment.

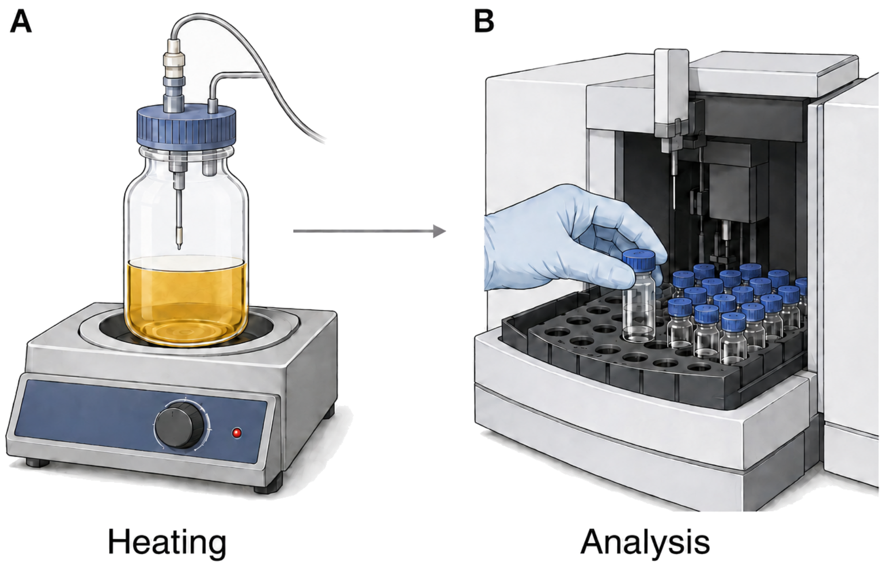

Figure 5. Heating and analysis of volatile products. A rapeseed-oil experiment measured volatile compounds formed during heating.[66] The illustration schematically shows heating and headspace sampling in panel A, followed by analytical sample handling in panel B. It does not reconstruct a particular instrument or encode compound quantities. Connecting such chemical measurements to human disease requires exposure and outcome evidence.[67]

Evidence for human disease remains less developed. Reviews identify oxidation products in abused frying oils as a plausible health concern.[68] A detailed review of aldehydic products calls for tolerable-intake values and direct epidemiological or nutritional studies.[67] An experimental study found tissue injury after rats consumed repeatedly heated oil.[69] That animal result supports hazard identification, not a quantitative estimate of risk from ordinary home cooking. Application to human dietary exposure requires information on oil identity, heating history, and the amounts of oxidation products consumed.

The assessment separates moderate confidence in altered oil chemistry from very low confidence in the magnitude of clinical harm at ordinary human exposure. Prospective studies measuring actual cooking exposures and subsequent health outcomes would help distinguish effects associated with oil identity from those associated with heating conditions.

13 Cancer, cognition, and inflammatory bowel disease

The cancer literature does not provide a single precise estimate for ordinary seed-oil exposure. A review concluded that high linoleic-acid intake was unlikely to substantially raise breast, colorectal, or prostate cancer risk, while acknowledging that a small increase could not be excluded.[70] A meta-analysis of 47 randomized trials found small point estimates with substantial uncertainty for cancer effects of total polyunsaturated fat, with sparse omega-6-specific evidence.[71] A newer global cohort synthesis associated higher dietary or circulating omega-6 with lower overall cardiovascular, cancer, and mortality risks, but reported heterogeneity by cancer site and health status.[72] The cancer conclusion remains limited because exposure definitions and sites differ.

Randomized-trial syntheses have also evaluated cognition and inflammatory bowel disease. A systematic review of randomized trials found that the effects of increasing omega-6 or total polyunsaturated fat on cognition were unclear.[73] Another randomized-trial synthesis found little support for modifying inflammatory bowel disease or long-term inflammatory status by increasing omega-3, omega-6, or total polyunsaturated fat.[74] The uncertainty for these outcomes should be retained separately from the findings for lipid and inflammatory biomarkers.

14 Conclusion

Replacing saturated fat with unsaturated oils lowers LDL cholesterol, with moderate confidence in the short-term lipid effects. Evidence for fewer coronary events is more limited and depends on the intervention; a mortality reduction is not consistently established. Prospective biomarker and cooking-oil associations do not identify the causal effects of individual dietary substitutions. Historical trials, contemporary feeding studies, and observational cohorts should therefore be interpreted according to their specific exposures and outcomes.

Reviews of common circulating inflammatory markers and a recent stimulated-blood experiment assess different biological responses. The latter demonstrates a biochemical effect without establishing a clinical disease consequence. Heating and storage studies document altered oil chemistry, but the magnitude of risk at ordinary human dietary exposures remains very uncertain. Long-duration studies of defined replacement diets and studies that measure heating-related exposure alongside health outcomes would address the principal limitations of the current evidence.

Scope and methods — Narrative review updated on 4 September 2026 using OpenAlex and PubMed, with ten search angles, review and primary-study searches, recent and contrary/null searches, and bidirectional citation chasing from six central sources. Human dietary interventions and observational findings were considered separately from laboratory and animal studies. This is not an exhaustive systematic review. Authenticated full texts and abstracts were used; full-text retrieval dates were preserved. Crossref screening found no integrity signal in the queried metadata for the included sources as of the check date; several publisher notice pages were inaccessible. Overlap between many reviews was not fully reconstructed and was not counted as independent corroboration. The certainty judgments are structured narrative assessments, not formal certainty ratings under the Grading of Recommendations Assessment, Development and Evaluation framework.

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74 · VERIFIED VIA CROSSREF · GROUNDED V0.5.2

107 CITED SENTENCES · 107 SOURCE CHECKS · 64 SUPPORTED AT FULL TEXT · 43 AT ABSTRACT · 0 PARTIAL · 0 CONTRADICTED

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